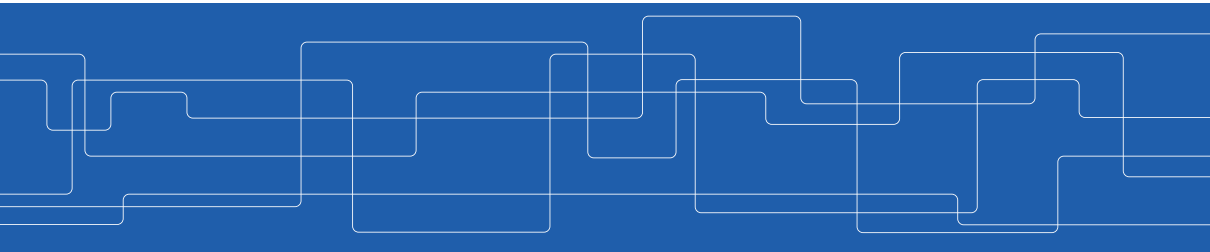


Regressions and Classification

Amir H. Payberah
payberah@kth.se
2026-10-30





Regression



Let's Start with an Example


FOR SALE
REALTY
CARTOONSTOCK.com
Search ID: rhan1292
THIS IS MY DREAM HOUSE ALRIGHT, EXCEPT IN MY DREAM IT WAS ABOUT HALF THIS PRICE.



The Housing Price Example (1/3)

- Given the dataset of `m` houses.

Living area	No. of bedrooms	Price
2104	3	400
1600	3	330
2400	3	369
⋮	⋮	⋮

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- Given the dataset of m houses.

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- Predict the prices of other houses, as a function of the size of living area and number of bedrooms?

The Housing Price Example (2/3)

Living area	No. of bedrooms	Price
2104	3	400
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Living area	No. of bedrooms	Price
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$$\mathbf{x}^{(1)} = \begin{bmatrix} 2104 \\ 3 \end{bmatrix} \quad \mathrm{y}^{(1)} = 400 \qquad \mathbf{x}^{(2)} = \begin{bmatrix} 1600 \\ 3 \end{bmatrix} \quad \mathrm{y}^{(2)} = 330 \qquad \mathbf{x}^{(3)} = \begin{bmatrix} 2400 \\ 3 \end{bmatrix} \quad \mathrm{y}^{(3)} = 369$$

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$$\mathbf{X} = \begin{bmatrix} \mathbf{x}^{(1)\mathsf{T}} \\ \mathbf{x}^{(2)\mathsf{T}} \\ \mathbf{x}^{(3)\mathsf{T}} \\ \vdots \end{bmatrix} = \begin{bmatrix} 2104 & 3 \\ 1600 & 3 \\ 2400 & 3 \\ \vdots & \vdots \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} 400 \\ 330 \\ 369 \\ \vdots \end{bmatrix}$$

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- $\mathbf{x}^{(i)} \in \mathbb{R}^2$: $x_1^{(i)}$ is the living area, and $x_2^{(i)}$ is the number of bedrooms of the ith house in the training set.

The Housing Price Example (3/3)

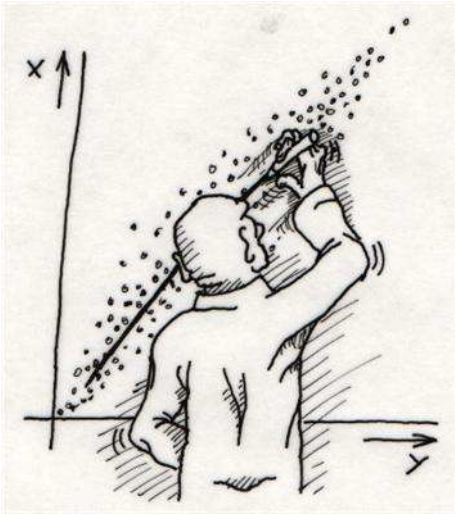
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- Predict the prices of other houses $\hat{y}$ as a function of the size of their living areas x_1, and number of bedrooms x_2, i.e., $\hat{y} = f(x_1, x_2)$
- E.g., what is $\hat{y}$, if $x_1 = 4000$ and $x_2 = 4$?

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- E.g., what is $\hat{y}$, if $x_1 = 4000$ and $x_2 = 4$?
- As an initial choice: $\hat{y} = f_w(\mathbf{x}) = w_1 x_1 + w_2 x_2$



[http://www.vias.org/science_cartoons/regression.html]



Linear Regression (1/2)

- Our goal: to build a system that takes input $\mathbf{x} \in \mathbb{R}^{\mathbf{n}}$ and predicts output $\hat{\mathbf{y}} \in \mathbb{R}$.

Linear Regression (1/2)

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- In linear regression, the output $\hat{y}$ is a linear function of the input $\mathbf{x}$.

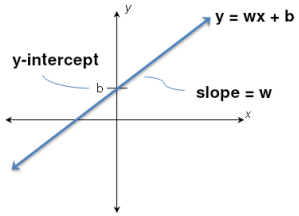
$$\hat{y} = f_w(\mathbf{x}) = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$
$$\hat{y} = \mathbf{w}^T \mathbf{x}$$

  - $\hat{y}$: the predicted value
  - n: the number of features
  - x_i: the ith feature value
  - w_j: the jth model parameter ($\mathbf{w} \in \mathbb{R}^n$)

Linear Regression (2/2)

- Linear regression often has one additional parameter, called intercept b:

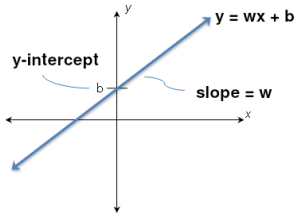
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Linear Regression (2/2)

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$$\hat{y} = \mathbf{w}^T\mathbf{x} + b$$



- Instead of adding the bias parameter b, we can augment $\mathbf{x}$ with an extra entry that is always set to 1.

$$\hat{y} = f_w(\mathbf{x}) = w_0x_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n, \text{ where } x_0 = 1$$



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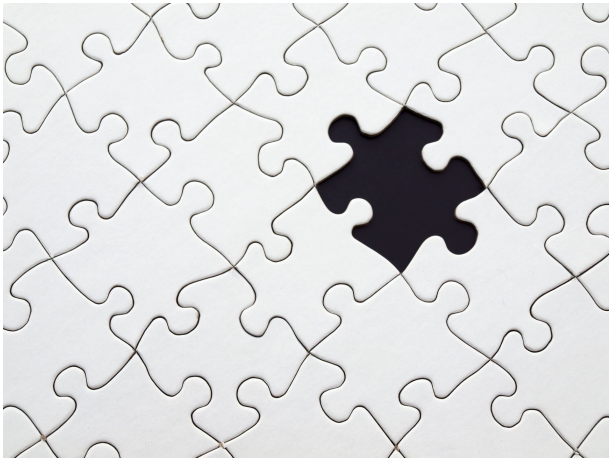


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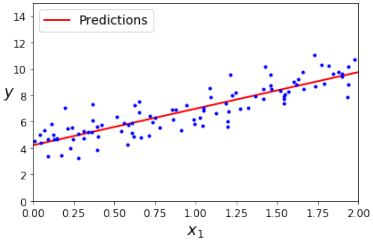
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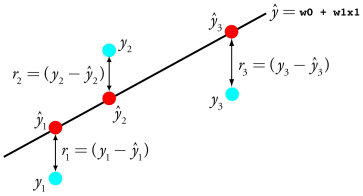


$$\hat{y} = f_w(\mathbf{x}) = w_0 x_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

How to Learn Model Parameters $\mathbf{w}$?

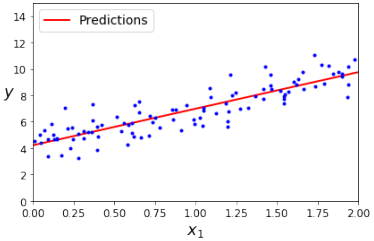
Linear Regression - Cost Function (1/2)

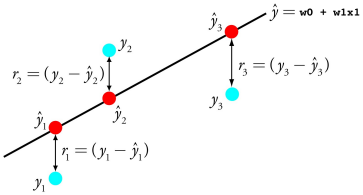




- One reasonable model should make $\hat{y}$ close to y, at least for the training dataset.

Linear Regression - Cost Function (1/2)

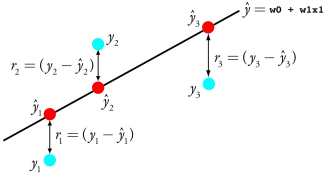




- One reasonable model should make $\hat{y}$ close to y, at least for the training dataset.
- Residual: the difference between the dependent variable y and the predicted value $\hat{y}$.

$$r^{(i)} = y^{(i)} - \hat{y}^{(i)}$$

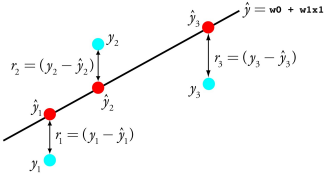
Linear Regression - Cost Function (2/2)



- Cost function $J(\mathbf{w})$
  - For each value of the $\mathbf{w}$, it measures how close the $\hat{y}^{(i)}$ is to the corresponding $y^{(i)}$.
  - We can define $J(\mathbf{w})$ as the mean squared error (MSE):

$$J(\mathbf{w}) = MSE(\mathbf{w}) = \frac{1}{m}\sum_{i}^{m}(\hat{y}^{(i)} - y^{(i)})^2$$

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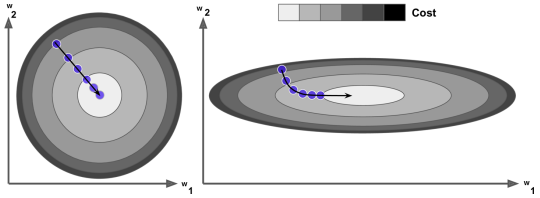
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$$= E[(\hat{y} - y)^2] = \frac{1}{m}||\hat{\mathbf{y}} - \mathbf{y}||_2^2$$

Gradient Descent (1/2)

- Gradient descent is a generic optimization algorithm capable of finding optimal solutions to a wide range of problems.
- The idea: to tweak parameters iteratively in order to minimize a cost function.



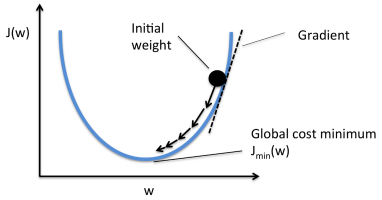
Gradient Descent (2/2)

- Suppose you are lost in the mountains in a dense fog.
- You can only feel the slope of the ground below your feet.
- A strategy to get to the bottom of the valley is to go downhill in the direction of the steepest slope.



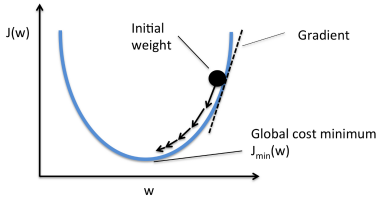
Gradient Descent - Iterative Optimization Algorithm

- Choose a starting point, e.g., filling **w** with random values.



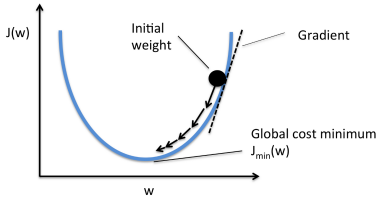
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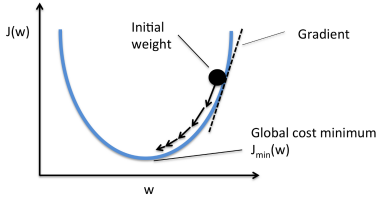
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- Determine the step size, the length of a step in the given direction.





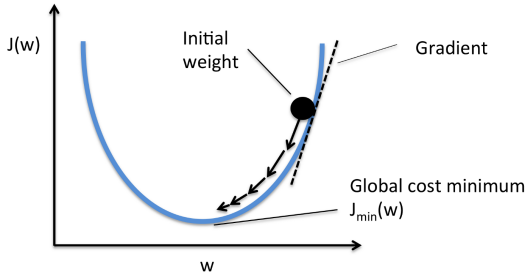
Gradient Descent - Key Points

- Stopping criterion
- Descent direction
- Step size (learning rate)

Gradient Descent - Stopping Criterion

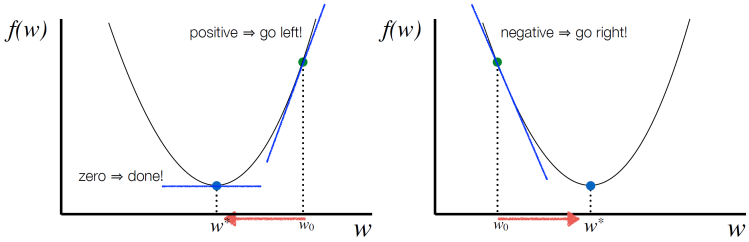
- The cost function minimum property: the gradient has to be zero.

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = 0$$



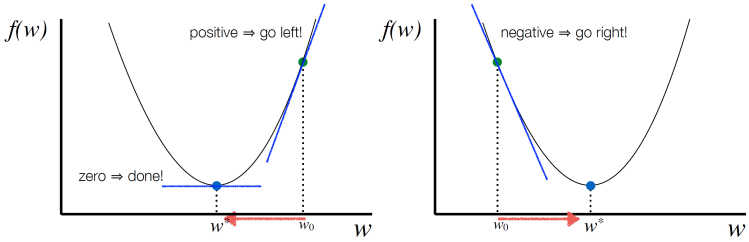
Gradient Descent - Descent Direction

- Direction in which the function value decreases near the current point.



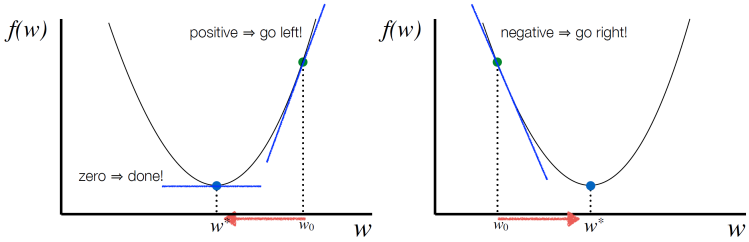
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- Find the direction of descent (slope).



Gradient Descent - Descent Direction

- Direction in which the function value decreases near the current point.
- Find the direction of descent (slope).
- Follow the opposite direction of the slope.



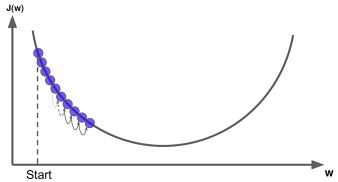


Gradient Descent - Learning Rate

- Learning rate: the length of steps.

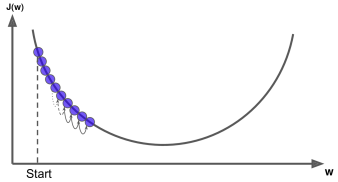
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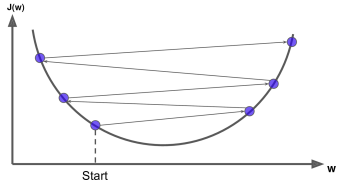
- Learning rate: the length of steps.
- If it is too small: many iterations to converge.



Gradient Descent - Learning Rate

- Learning rate: the length of steps.
- If it is too small: many iterations to converge.
- If it is too high: the algorithm might diverge.





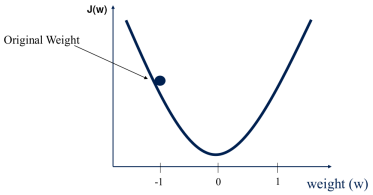


Gradient Descent - How to Learn Model Parameters $\mathbf{w}$?

- Goal: find $\mathbf{w}$ that minimizes $J(\mathbf{w}) = \sum_{i=1}^{m} (\mathbf{w}^T \mathbf{x}^{(i)} - y^{(i)})^2$.

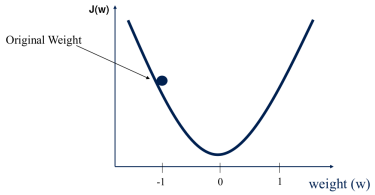
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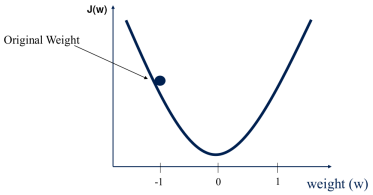
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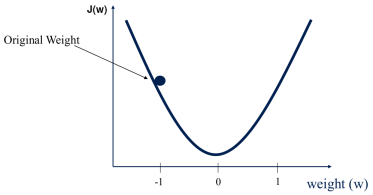
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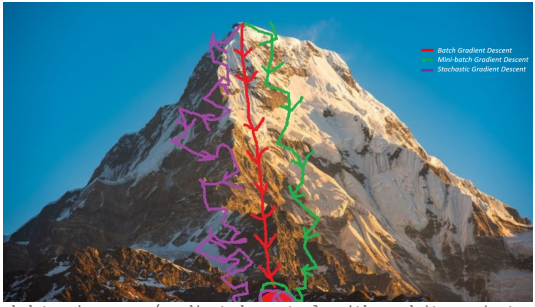
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  1. Determine a descent direction $\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}}$
  2. Choose a step size η
  3. Update the parameters: $w^{(next)} = w - \eta \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}}$
     (should be done for all parameters simultanously)



Gradient Descent - Different Algorithms (1/3)

- Batch gradient descent
- Stochastic gradient descent
- Mini-batch gradient descent



[https://towardsdatascience.com/gradient-descent-algorithm-and-its-variants-10f652806a3]



Gradient Descent - Different Algorithms (2/3)

- Batch gradient descent: at each step, it computes the gradients based on the full training set.



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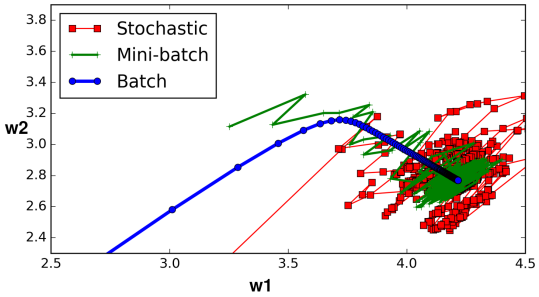


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- Mini-batch gradient descent: at each step, it computes the gradients based on small random sets of instances called mini-batches.

Gradient Descent - Different Algorithms (3/3)

Algorithm	Large m	Large n
Normal Equation	Fast	Slow
Batch GD	Slow	Fast
Stochastic GD	Fast	Fast
Mini-batch GD	Fast	Fast





Gradient Descent - Different Algorithms (2/3)

- Batch gradient descent: at each step, it computes the gradients based on the full training set.



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Accuracy vs. Fariness (1/2)

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Accuracy vs. Fariness (1/2)

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- The model achieves a very low MSE, but it was trained mostly on data from well-resourced schools.
- As a result, it systematically underpredicts the performance of students in under-resourced areas, leading to fewer resources where they are needed most.
- Question: Should we still value a lower MSE if achieving it means reinforcing existing inequalities and causing harm to certain groups?



Accuracy vs. Fariness (2/2)

- We should ask: Better for whom?



Accuracy vs. Fariness (2/2)

- We should ask: Better for whom?
- MSE minimizes total error, not fairness: marginalized groups may carry more error.



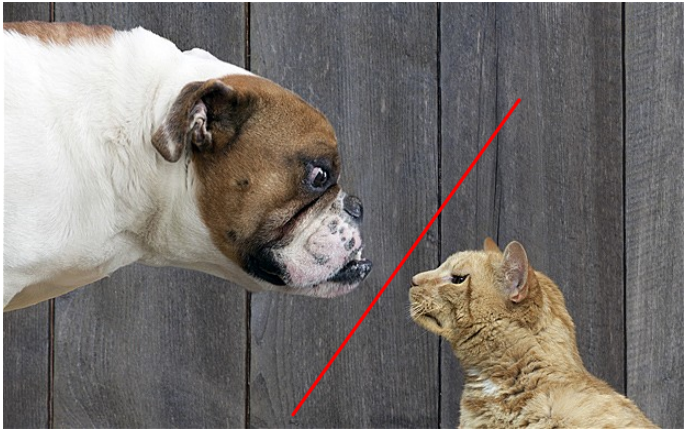
Accuracy vs. Fariness (2/2)

- We should ask: Better for whom?
- MSE minimizes total error, not fairness: marginalized groups may carry more error.
- Explore alternative metrics balancing equity and performance (e.g., group-weighted loss).



Classification

Let's Start with an Example



[https://www.telegraph.co.uk/lifestyle/pets/8151921/Dogs-are-smarter-than-cats-feline-friends-disagree.html]



Example (1/4)

- Given the dataset of m cancer tests.

Tumor size	Cancer
330	1
120	0
400	1
⋮	⋮



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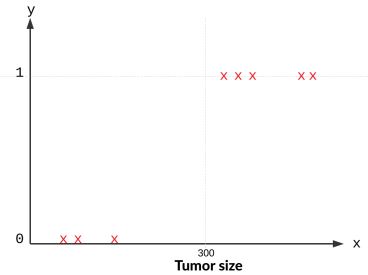
Tumor size	Cancer
330	1
120	0
400	1
⋮	⋮

- Predict the risk of cancer, as a function of the tumor size?

Example (2/4)

Tumor size	Cancer
330	1
120	0
400	1
⋮	⋮

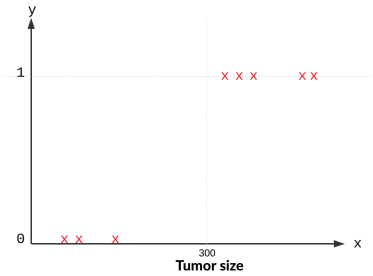
$$\mathbf{x} = \begin{bmatrix} 330 \\ 120 \\ 400 \\ \vdots \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ \vdots \end{bmatrix}$$



Example (2/4)

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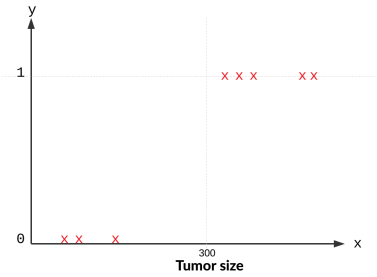


- $\mathbf{x}^{(i)} \in \mathbb{R}$: $x_1^{(i)}$ is the tumor size of the ith instance in the training set.

Example (3/4)

Tumor size	Cancer
330	1
120	0
400	1
⋮	⋮

$$\mathbf{x} = \begin{bmatrix} 330 \\ 120 \\ 400 \\ \vdots \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ \vdots \end{bmatrix}$$

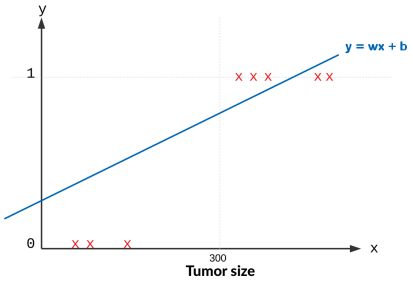


- Predict the risk of cancer $\hat{y}$ as a function of the tumor sizes x_1, i.e., $\hat{y} = f(x_1)$
- E.g., what is $\hat{y}$, if $x_1 = 500$?

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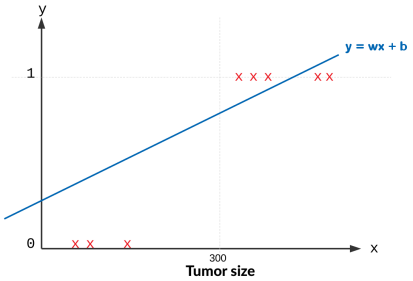


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Example (3/4)

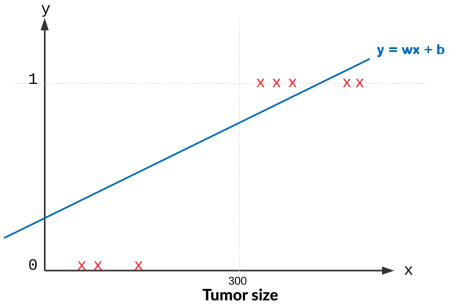
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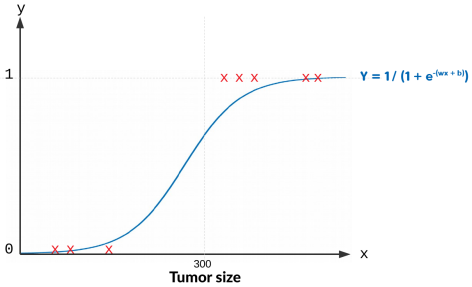
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- E.g., what is $\hat{y}$, if $x_1 = 500$?
- As an initial choice: $\hat{y} = f_w(\mathbf{x}) = w_0 + w_1 x_1$
- Bad model!

Example (4/4)



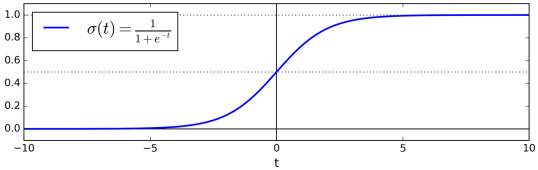


- A better model $\hat{y} = \frac{1}{1+e^{-(w_0+w_1x_1)}}$

Sigmoid Function

- The sigmoid function, denoted by $\sigma(.)$, outputs a number between 0 and 1.

$$\sigma(t) = \frac{1}{1 + e^{-t}}$$

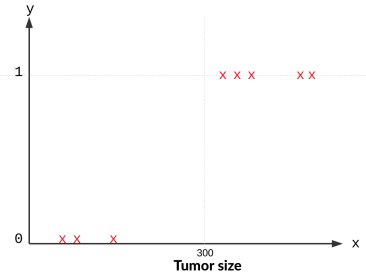


- When $t < 0$, then $\sigma(t) < 0.5$
- when $t \geq 0$, then $\sigma(t) \geq 0.5$

Binomial Logistic Regression

Binomial Logistic Regression (1/2)

- Our goal: to build a system that takes input $\mathbf{x} \in \mathbb{R}^n$ and predicts output $\hat{y} \in \{0, 1\}$.
- To specify which of 2 categories an input $\mathbf{x}$ belongs to.





Binomial Logistic Regression (2/2)

- Linear regression: the model computes the weighted sum of the input features (plus a bias term).

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n = \mathbf{w}^T \mathbf{x}$$

Binomial Logistic Regression (2/2)

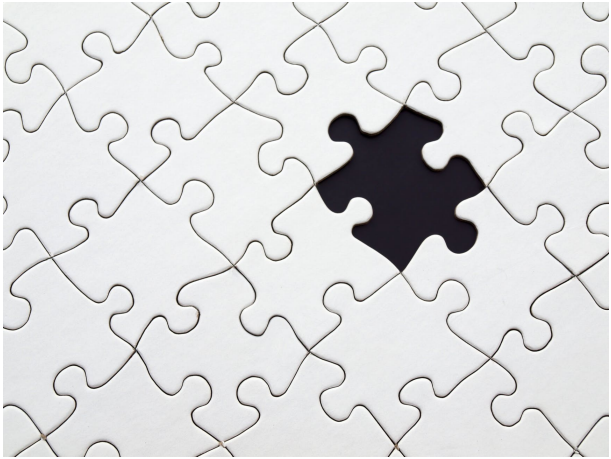
- Linear regression: the model computes the weighted sum of the input features (plus a bias term).

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n = \mathbf{w}^\intercal \mathbf{x}$$

- Binomial logistic regression: the model computes a weighted sum of the input features (plus a bias term), but it outputs the logistic of this result.

$$z = w_0 x_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n = \mathbf{w}^\intercal \mathbf{x}$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-\mathbf{w}^\intercal \mathbf{x}}}$$



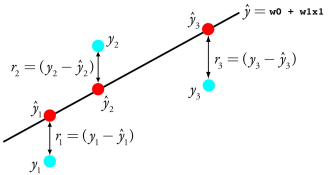
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How to Learn Model Parameters $\mathbf{w}$?

Linear Regression - Cost Function



- One reasonable model should make $\hat{y}$ close to y, at least for the training dataset.
- Cost function $J(\mathbf{w})$: the mean squared error (MSE)

$$\text{cost}(\hat{y}^{(i)}, y^{(i)}) = (\hat{y}^{(i)} - y^{(i)})^2$$

$$J(\mathbf{w}) = \frac{1}{m}\sum_{i}^{m} \text{cost}(\hat{y}^{(i)}, y^{(i)}) = \frac{1}{m}\sum_{i}^{m} (\hat{y}^{(i)} - y^{(i)})^2$$



Binomial Logistic Regression - Cost Function (1/5)

- Naive idea: minimizing the Mean Squared Error (MSE)

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$$J(\mathbf{w}) = \text{MSE}(\mathbf{w}) = \frac{1}{m}\sum_{i}^{m} \left(\frac{1}{1 + e^{-\mathbf{w}^T\mathbf{x}^{(i)}}} - y^{(i)}\right)^2$$

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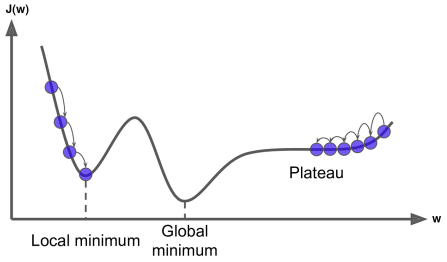
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- This cost function is a non-convex function for parameter optimization.

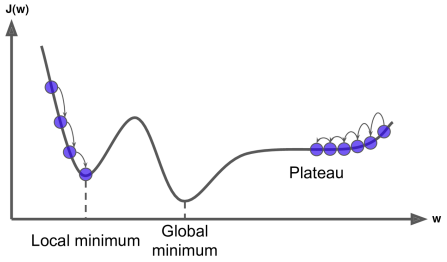
Binomial Logistic Regression - Cost Function (2/5)

- What do we mean by non-convex?
- If a line joining two points on the curve, crosses the curve.
- The algorithm may converge to a local minimum.



Binomial Logistic Regression - Cost Function (2/5)

- What do we mean by non-convex?
- If a line joining two points on the curve, crosses the curve.
- The algorithm may converge to a local minimum.
- We want a convex logistic regression cost function $J(\mathbf{w})$.





Binomial Logistic Regression - Cost Function (3/5)

- The predicted value $\hat{y} = \sigma(\mathbf{w}^\intercal \mathbf{x}) = \frac{1}{1+e^{-\mathbf{w}^\intercal \mathbf{x}}}$
- $\mathtt{cost}(\hat{y}^{(i)}, y^{(i)}) = ?$

Binomial Logistic Regression - Cost Function (3/5)

- The predicted value $\hat{y} = \sigma(\mathbf{w}^T\mathbf{x}) = \frac{1}{1+e^{-\mathbf{w}^T\mathbf{x}}}$
- $\mathtt{cost}(\hat{y}^{(i)}, y^{(i)}) = ?$
- The $\mathtt{cost}(\hat{y}^{(i)}, y^{(i)})$ should be
  - Close to 0, if the predicted value $\hat{y}$ will be close to true value y.
  - Large, if the predicted value $\hat{y}$ will be far from the true value y.

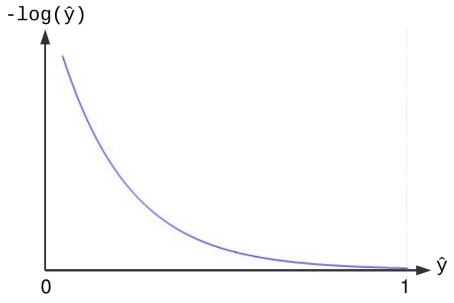
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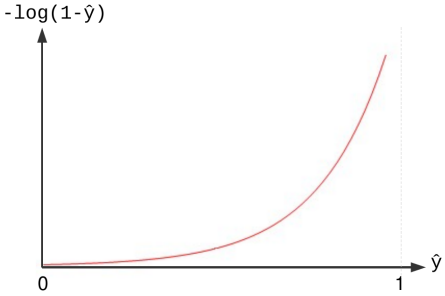
$$\mathtt{cost}(\hat{y}^{(i)}, y^{(i)}) = \begin{cases} -\log(\hat{y}^{(i)}) & \text{if } y^{(i)} = 1 \\ -\log(1-\hat{y}^{(i)}) & \text{if } y^{(i)} = 0 \end{cases}$$

Binomial Logistic Regression - Cost Function (4/5)

$$\texttt{cost}(\hat{\texttt{y}}^{(\texttt{i})}, \texttt{y}^{(\texttt{i})}) = \begin{cases} -\texttt{log}(\hat{\texttt{y}}^{(\texttt{i})}) & \text{if} \quad \texttt{y}^{(\texttt{i})} = 1 \\ -\texttt{log}(1 - \hat{\texttt{y}}^{(\texttt{i})}) & \text{if} \quad \texttt{y}^{(\texttt{i})} = 0 \end{cases}$$



when $\texttt{y} = 1$



when $\texttt{y} = 0$

Binomial Logistic Regression - Cost Function (5/5)

- We can define $\mathtt{J}(\mathbf{w})$ as below

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Binomial Logistic Regression - Cost Function (5/5)

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$$J(\mathbf{w}) = \frac{1}{m} \sum_{i}^{m} \text{cost}(\hat{y}^{(i)}, y^{(i)}) = -\frac{1}{m} \sum_{i}^{m} (y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}))$$



Binomial Logistic Regression Gradient Descent

- Goal: find $\mathbf{w}$ that minimizes $J(\mathbf{w}) = -\frac{1}{m}\sum_i^m (y^{(i)}\log(\hat{y}^{(i)}) + (1-y^{(i)})\log(1-\hat{y}^{(i)}))$.



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  1. Determine a descent direction $\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}}$



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- Start at a random point, and repeat the following steps, until the stopping criterion is satisfied:
  1. Determine a descent direction $\frac{\partial J(\mathbf{w})}{\partial w}$
  2. Choose a step size η
  3. Update the parameters: $w^{(next)} = w - \eta \frac{\partial J(\mathbf{w})}{\partial w}$ (simultaneously for all parameters)

Multinomial Logistic Regression



Multinomial Logistic Regression

- Multinomial classifiers can distinguish between more than two classes.
- Instead of $\mathtt{y} \in \{0, 1\}$, we have $\mathtt{y} \in \{1, 2, \cdots, \mathtt{k}\}$.



Binomial vs. Multinomial Logistic Regression

- In a binary class, $y \in \{0, 1\}$, we use the sigmoid function.

$$\hat{y} = p(y = 1 \mid \mathbf{x}; \mathbf{w}) = \sigma(\mathbf{w}^T \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}}$$



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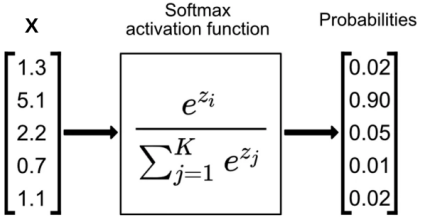
$$\hat{y} = p(y = 1 \mid \mathbf{x}; \mathbf{w}) = \sigma(\mathbf{w}^\intercal \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\intercal \mathbf{x}}}$$

- In multiclasses, $y \in \{1, 2, \cdots, k\}$, we use the softmax function.

$$\hat{y}_j = p(y = j \mid \mathbf{x}; \mathbf{w}_j) = \sigma(\mathbf{w}_j^\intercal \mathbf{x}) = \frac{e^{\mathbf{w}_j^\intercal \mathbf{x}}}{\sum_{i=1}^{k} e^{\mathbf{w}_i^\intercal \mathbf{x}}}$$

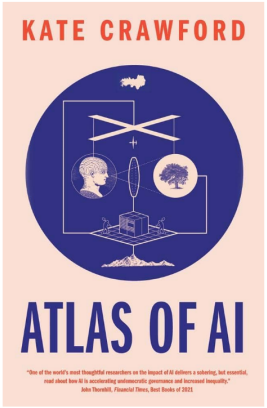
Sigmoid vs. Softmax

- Sigmoid function: $\sigma(\mathbf{w}^\intercal \mathbf{x}) = \frac{1}{1+e^{-\mathbf{w}^\intercal \mathbf{x}}}$
- Softmax function: $\sigma(\mathbf{w}_j^\intercal \mathbf{x}) = \frac{e^{\mathbf{w}_j^\intercal \mathbf{x}}}{\sum_{i=1}^{k} e^{\mathbf{w}_i^\intercal \mathbf{x}}}$
  - Calculate the probabilities of each target class over all possible target classes.
  - The softmax function for two classes is equivalent the sigmoid function.





Atlas of AI





https://tinyurl.com/97spajab

Samuel Morton Skull Collection

- Collected over 1000 skulls in the early 1800s.
- Skull labels include racial and descriptive terms.
- Aimed to classify and rank human races objectively.



https://tinyurl.com/y7mn963k

Morton's Classification

- Used cranial measurements to argue for racial hierarchy in intelligence.
- Categorized into five races:
  - African
  - Native American
  - Caucasian
  - Malay
  - Mongolian



https://tinyurl.com/2dfv2fd3

Scientific Impact and Criticism

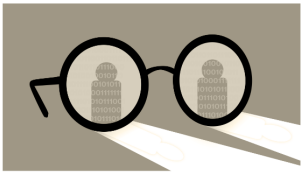
- Considered cutting-edge science of the time.
- Cited for nearly a century to justify racial hierarchies.
- Used to legitimize slavery and racial segregation in the US.



https://tinyurl.com/z9syrybx

Reassessment of Morton's Work

- Found selective sampling, procedural omissions, and assumptions that influenced results.
- Emphasized the social context influencing Morton's findings.
- Classification systems in AI can perpetuate biases.
- Importance of questioning underlying assumptions in data classification.



https://tinyurl.com/9nmkws7k



Questions Raised

- Who creates and changes these categories?
- What values might be encoded into them?
- What hidden hierarchies they encode?
- How do they spread and affect public policy?
- Whether these classifications should exist in the first place?

Power and Ethics of Classifications (1/2)

- Segmentation of the world; set of boxes (metaphorical or literal) to organize items for work or knowledge production.

- Embedded in our daily life.
  - Sorting clothes for the wash.
  - Classifying medications and books.
  - Organizing folders in computers.



https://tinyurl.com/k89s9aab

Power and Ethics of Classifications (2/2)

- These systems are usually invisible but crucial for societal order.
- They favor some perspectives while silencing others.
- When classifications fail, they become noticeable.
- Examples
  - Airport security systems (body scanners)
  - Automated hiring tools
  - Washed mixed color clothes



https://tinyurl.com/y653brh5

Summary



Summary

- Linear regression
  - Learning parameters $\mathbf{w}$
  - Cost function $J(\mathbf{w})$
  - Learn parameters: normal equation, gradient descent (batch, stochastic, mini-batch)
- Classification
  - Sigmoid vs. Softmax
- Ciritical view



Reference

- Hands-on machine learning with Scikit-Learn and TensorFlow (Ch. 3, 4)
- Atlas of AI (Ch. 4)



Questions?